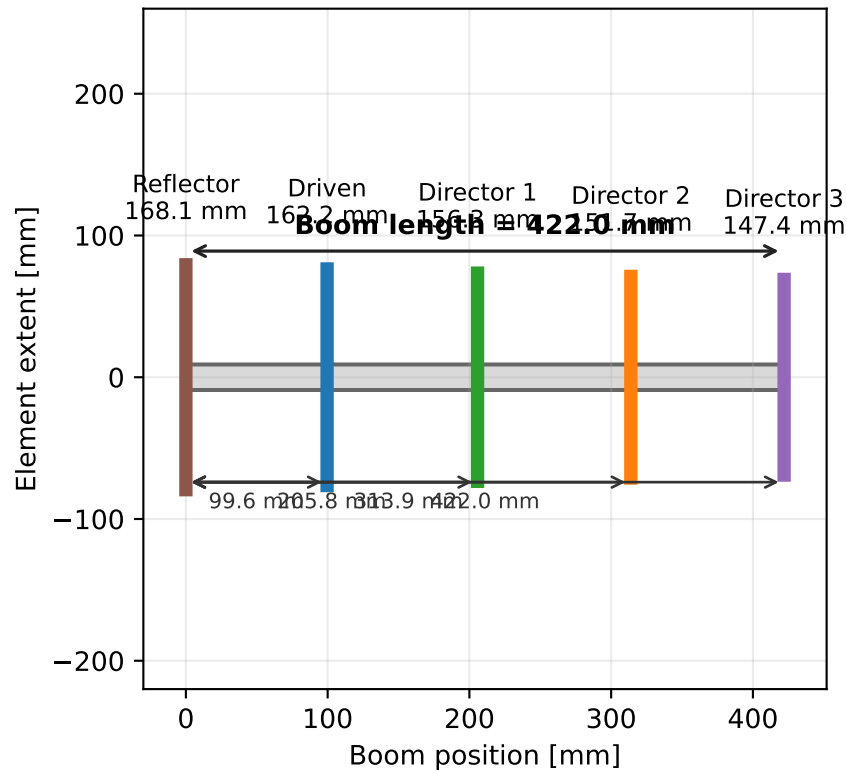


Yagi-Uda 915 MHz Design Summary (Balanis 3rd ed.)

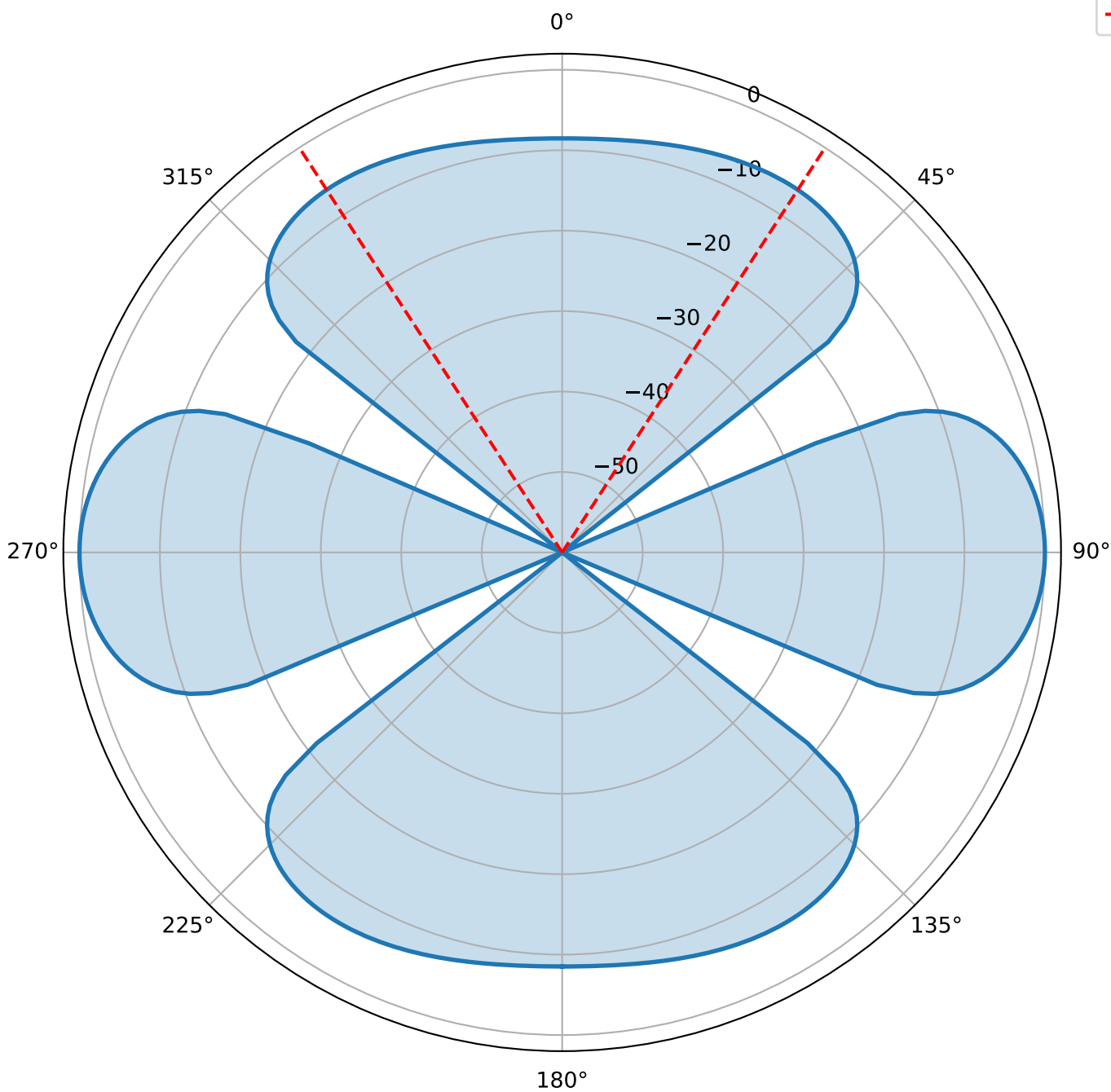
Parameter	Value
Operating frequency	915.000 MHz
Free space wavelength	327.64 mm
Wavenumber k	19.177 rad/m
Reflector length	168.08 mm
Driven element length	162.18 mm
Director 1 length	156.29 mm
Director 2 length	151.70 mm
Director 3 length	147.44 mm
Reflector position	0.00 mm
Driven position	99.60 mm
Director 1 position	205.76 mm
Director 2 position	313.88 mm
Director 3 position	422.00 mm
Total boom length	422.00 mm
Boom length / wavelength	1.288 λ
Half-wave dipole Z (real)	73.00 ohm
Half-wave dipole Z (imag)	42.50 ohm
Radiation resistance	73.00 ohm
Directivity (from Fig. 10.14)	11.10 dBi
Directivity (linear)	12.88 x
Radiation efficiency	98.0 %
Gain (directivity $\times$ efficiency)	11.01 dBi
Gain (linear)	12.62 x
Driven element impedance	25.00 ohm
System impedance	50.00 ohm
Reflection coefficient	0.333 unitless
VSWR into 50 ohm system	2.000:1
Return loss	9.54 dB
Mismatch loss	0.512 dB
E-plane HPBW	33.00 deg
Front-to-back ratio	18.00 dB
Front-to-back power ratio	63.10 x
Effective aperture	1100.50 cm ²
Hairpin match length	20.48 mm

5-Element Yagi-Uda Element Layout at 915 MHz

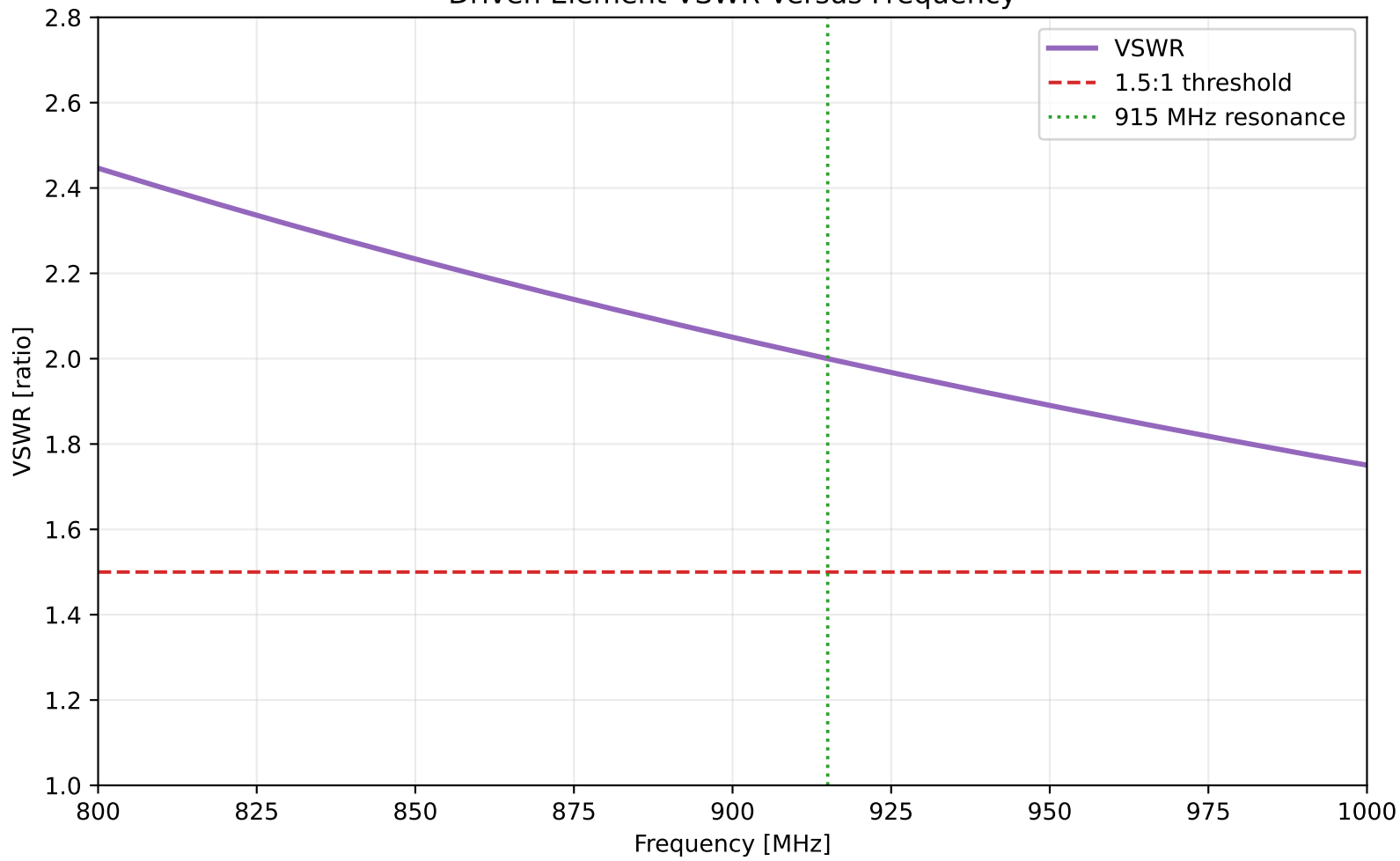


E-Plane Normalised Array Factor Pattern at 915 MHz
(Balanis 3rd ed., Ch. 6, Sec. 6.2)

— Array Factor
- - - HPBW = $\pm 33.0^\circ$



Driven Element VSWR Versus Frequency



Gain Versus Number of Elements
(Balanis 3rd ed., Ch. 10, Table 10.6)

